

Sharp interface limit of the Allen–Cahn equation with small periodic heterogeneities

Joint work with R. Cristoferi (Radboud) and L. Ganedi (Utah)

Luca Pignatelli (luca.pignatelli@ru.nl)

Setup of the problem

Let $\Omega \subset \mathbb{R}^d$ ($d \geq 2$) open and bounded with C^2 boundary, $\Theta: \Omega \rightarrow (0, \infty)$ a Q -periodic L^2 -function with $1/c \leq \Theta \leq c$, $Q = (-\frac{1}{2}, \frac{1}{2})^d$, and $W \in C^2(\mathbb{R}; [0, \infty))$ a double-well potential vanishing at $\{a, b\}$ with linear growth from below at infinity. For $\varepsilon, \delta > 0$, consider

$$F_{\varepsilon, \delta}(u) := \int_{\Omega} \left[\frac{1}{\varepsilon} \Theta \left(\frac{x}{\delta} \right) W(u) + \frac{\varepsilon}{2} |\nabla u|^2 \right] dx.$$

The associated (time-rescaled) heterogeneous Allen–Cahn L^2 -gradient flow is

$$\partial_t u_{\varepsilon, \delta} = \Delta u_{\varepsilon, \delta} - \frac{1}{\varepsilon^2} \Theta \left(\frac{x}{\delta} \right) W'(u_{\varepsilon, \delta}), \quad \partial_{\nu} u_{\varepsilon, \delta} = 0 \text{ on } \partial\Omega.$$

In the regime $\delta \ll \varepsilon$, $F_{\varepsilon, \delta}$ Γ -converges to a multiple of the perimeter (see [1])

$$F_0^h(u) = \sigma^h \text{Per}(\{u = a\}; \Omega), \quad \sigma^h = \sqrt{\Theta^h} \int_a^b \sqrt{2W(s)} ds, \quad \Theta^h = \int_Q \Theta(y) dy.$$

Question. Do the gradient flows converge to a (weak) Mean–Curvature Flow (MCF) of F_0^h , i.e. $V = -\sigma^h H$ on the moving interface, when $\varepsilon, \delta \rightarrow 0$ jointly?

Where we sit in the literature

- **Feldman–Morfe** [2] and similar studies: $\partial_t u_{\varepsilon, \delta} = \Delta u_{\varepsilon, \delta} - \frac{1}{\varepsilon^2} \Theta \left(\frac{x}{\varepsilon} \right) W'(u_{\varepsilon, \delta})$. Using a *level-set* / *viscosity solution* formulation they showed that pathologies are the norm: gradient discontinuities of the effective surface tension, bubbling, and *interface pinning*.
- **Ganedi–Marveggio–Stinson** [3]: $\partial_t u_{\varepsilon, \delta} = \Delta u_{\varepsilon, \delta} - \frac{1}{\varepsilon^2} \partial_u W(x, u_{\varepsilon, \delta})$ with $a(x), b(x) \in C^2(\Omega)$ separated wells. They use *BV solutions* and show convergence to a *weighted MCF* (the limit equation carries the heterogeneity).
- **This work.** Multiplicative potential $\Theta(x/\delta)W(u)$ **with two scales ε and δ** . The oscillations *average out* in the limit, so the resulting motion law is the *isotropic* MCF $V = -\sigma^h H$ for the homogenized surface tension σ^h — **no** pathologies, **no** pinning, in the regime $\delta \ll \varepsilon^2$.

Weak notion of MCF used

We use **BV solutions** (Luckhaus–Sturzenhecker [4], Laux–Simon [5]): purely variational, naturally tied to the energy–dissipation inequality, well-suited to handle the limit of the gradient flow.

Definition. Fix $T > 0$ and $A_0 \subset \Omega$ of finite perimeter. A family $\{A(t)\}_{t \in [0, T]}$ is a *BV solution to MCF of F_0^h* with initial datum A_0 if

- $\mathbb{1}_{A(\cdot)} \in C^{1/2}([0, T]; L^1(\Omega)) \cap \text{BV}(\Omega \times (0, T))$ and $A(0) = A_0$ in $L^1(\Omega)$;
- there exists $V \in L^2(|\nabla \mathbb{1}_{A(\cdot)}| \otimes \mathcal{L}^N_{(0, T)})$ such that, for a.e. $S \in (0, T)$ and all $\phi \in C_c^\infty(\bar{\Omega} \times [0, S))$,

$$\int_{A(S)} \phi(\cdot, S) dx - \int_{A(0)} \phi(\cdot, 0) dx = \int_0^S \int_{A(t)} \partial_t \phi dx dt - \int_0^S \int_{\Omega} V \phi d|\nabla \mathbb{1}_{A(t)}| dt;$$

- for a.e. $t \in (0, T)$ and all $\phi \in C^1(\bar{\Omega}; \mathbb{R}^d)$ with $\phi \cdot \nu_{\partial\Omega} = 0$,

$$\int_{\partial^* A(t)} V(\phi \cdot \nu_{\partial A(t)}) d\mathcal{H}^{N-1} = - \int_{\partial^* A(t)} \sigma^h [(\text{Id} - \nu_{\partial A(t)} \otimes \nu_{\partial A(t)}) : \nabla \phi] d\mathcal{H}^{N-1}.$$

The third condition is the *weak motion law* $V = -\sigma^h H_{\partial^* A(t)}$.

Ongoing and future work

- Is $\delta \ll \varepsilon^2$ sharp? In the intermediate regime $\varepsilon^2 \ll \delta \ll \varepsilon$ numerics suggest that oscillation-induced *pinning* effects take over and the limit may cease to be a BV solution.
- Extensions to general potentials $W(x/\delta, u)$, vector-valued / multiphase, mass-constrained, multi-scale.
- What about $\varepsilon \ll \delta$? Homogenization of weighted MCF?

Selected references

- [1] Riccardo Cristoferi, Irene Fonseca, and Likhit Ganedi. “Homogenization and Phase Separation with Fixed Wells—The Supercritical Case”. In: *SIAM Journal on Mathematical Analysis* 57.2 (2025), pp. 2138–2173.
- [2] William M Feldman and Peter Morfe. “The occurrence of surface tension gradient discontinuities and zero mobility for Allen–Cahn and curvature flows in periodic media”. In: *Interfaces and Free Boundaries* 25.4 (2023), pp. 567–631.
- [3] L. Ganedi, A. Marveggio, and K. Stinson. “Convergence of the heterogeneous Allen–Cahn equation to weighted mean curvature flow”. In: *Preprint* (2025).
- [4] S. Luckhaus and T. Sturzenhecker. “Implicit time discretization for the mean curvature flow equation”. In: *Calc. Var. Partial Differential Equations* 3 (1995), pp. 253–271.
- [5] T. Laux and T. M. Simon. “Convergence of the Allen–Cahn equation to multiphase mean curvature flow”. In: *Comm. Pure Appl. Math.* 71 (2018), pp. 1597–1647.

Main results

Theorem (Pre-compactness, $\delta_n \ll \varepsilon_n$). Let $\{u_n^0\}_n \subset L^\infty(\Omega)$ with $\sup_n \|u_n^0\|_\infty < \infty$ and $\sup_n F_{\varepsilon_n, \delta_n}(u_n^0) < \infty$, and let u_n be the (unique) weak solution to the heterogeneous AC equation with initial datum u_n^0 . Then there is a subsequence and a family $\{A(t)\}_{t \in (0, T)} \subset \text{BV}(\Omega; \{a, b\})$ with

$$u_{n_k} \rightarrow \mathbb{1}_{A(\cdot)} \quad \text{strongly in } L^1(\Omega \times (0, T)).$$

Theorem (Convergence to MCF, $\delta_n \ll \varepsilon_n^2$). Under the same setting, assume the initial data are well-prepared, i.e. $u_n^0 \rightarrow \mathbb{1}_{A_0}$ in $L^1(\Omega)$ with $\lim_n F_{\varepsilon_n, \delta_n}(u_n^0) = F_0^h(\mathbb{1}_{A_0})$, and that the time-averaged energies converge:

$$\lim_{n \rightarrow \infty} \int_0^T F_{\varepsilon_n, \delta_n}(u_n(\cdot, t)) dt = \int_0^T F_0^h(\mathbb{1}_{A(t)}) dt.$$

Then $\{A(t)\}_{t \in (0, T)}$ is a **BV solution to MCF of F_0^h** with initial datum A_0 .

Two key ingredients. *Asymptotic equipartition* ($\delta_n \ll \varepsilon_n$) gives that gradient and potential energies equidistribute on $\sigma^h |\nabla \mathbb{1}_A| \llcorner \Omega$.

Asymptotic dissipation ($\delta_n \ll \varepsilon_n^2$) says that the oscillations of $\Theta(\cdot/\delta_n)$ in the L^2 -gradient $\nabla F_{\varepsilon_n, \delta_n}(u_n)$ are negligible:

$$\lim_n \int_0^T \int_{\Omega} \nabla F_{\varepsilon_n, \delta_n}(u_n) \cdot (\nabla u_n \cdot \psi) dx dt = - \int_0^T \int_{\Omega} (\psi \cdot \nu) \sigma^h H_{A(t)} d|\nabla \mathbb{1}_{A(t)}| dt.$$

Why these assumptions?

- *Multiplicative potential.* The equipartition lemma estimates

$$\Delta_n := F_{\varepsilon_n, \delta_n}(u_n) - F_{\varepsilon_n}^h(u_n) = \int_{\Omega} \frac{1}{\varepsilon_n} [\Theta(\frac{x}{\delta_n}) - \Theta^h] W(u_n) dx$$

using $\text{div}(f(y)) = \Theta(y) - \Theta^h$, integrating by parts and bounding $W'(z) \leq C\sqrt{W(z)}$, to obtain $\sqrt{W(u_n)} |\nabla u_n|$, uniformly bounded in L^1 by compactness.

- *Regime.* In the dissipation analysis the worst term is $\frac{\delta_n}{\varepsilon_n} \int_{\Omega} |\nabla[W'(u_n) \nabla u_n \cdot \psi]| dx \lesssim \frac{\delta_n}{\varepsilon_n} \|W'(u_n)\|_{L^2} \|\nabla^2 u_n\|_{L^2}$, with

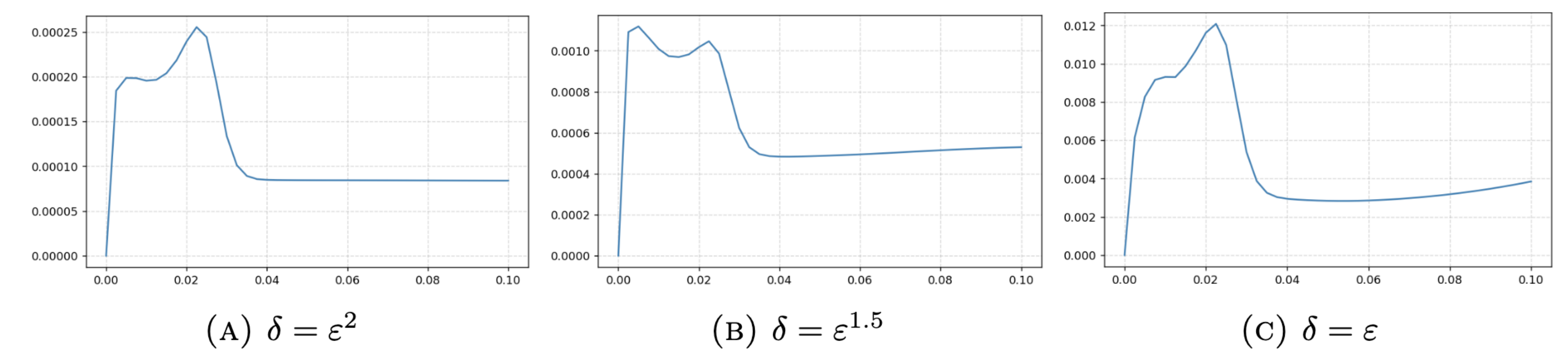
$$\|\nabla^2 u_n\|_{L^2} \lesssim \|\partial_t u_n\|_{L^2} + \frac{1}{\varepsilon_n^2} \left\| \Theta(\frac{\cdot}{\delta_n}) W'(u_n) \right\|_{L^2} \lesssim \frac{1}{\varepsilon_n^2} \|W'(u_n)\|_{L^2}$$

by elliptic regularity, energy-dissipation and the L^∞ bound on u_n , yielding δ_n/ε_n^3 . An order of ε_n is recovered thanks to $W'(z) \leq C\sqrt{W(z)}$.

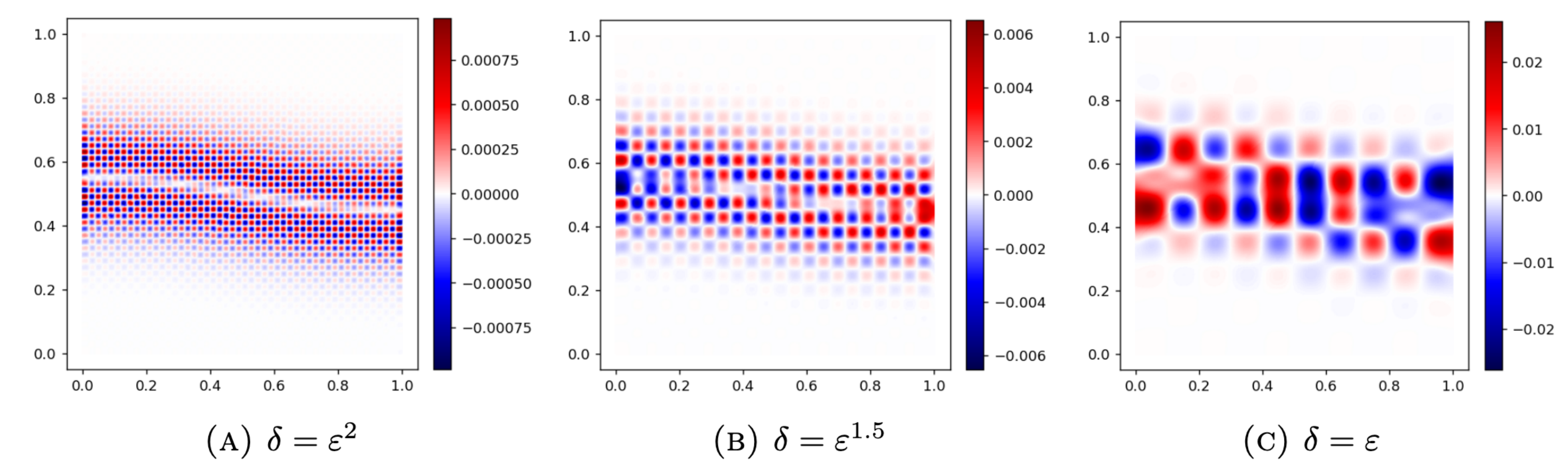
Numerics

$\Omega = [0, 1]^2$, $W(u) = (1 - u^2)^2$, $\Theta(y) = \sin(2\pi y_1) \sin(2\pi y_2) + 2$, $\Theta^h = 2$. P^1 FE on a uniform triangular mesh ($h = 1/500$), Shen–Yang stabilised semi-implicit scheme with $\Delta t = 5 \times 10^{-4}$, $T = 0.1$. Sinusoidal initial datum (pinned at $x = 0, 1$), $\varepsilon = 0.2$ fixed.

L^1 difference between evolutions over time:



Pointwise differences between evolutions at $t = T/2$:



Pointwise differences between evolutions at $t = T$:

